Advanced Algorithms 2025 2025

Sample Solutions GHW 02

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1 Server Management

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1. Consider an arbitrary $s \in \mathcal{S}$ and its point in the unit circle θ_s. Now consider the part of the circle of arc length $x \in (0, 2\pi)$ starting clockwise from θ_s. Notice that since the random variables $\{\theta_i\}_{i\in[n]}$ are fully independent, the aforementioned region is empty of any other server points with probability $\left(1 - \frac{x}{2\pi}\right)^{n-1}$. In other words, if Z_s is the maximal arc that is empty of server points (other than θ_s) starting clockwise from θ_s, then $\mathbb{P}[Z_s \geq x] = \left(1 - \frac{x}{2\pi}\right)^{n-1}$. Consequently, since Z_s is a non-negative random variable, we have

$$\mathbb{E}[Z_s] = \int_{z=0}^{\infty} \mathbb{P}[Z_s \geq z]dz = \int_{z=0}^{2\pi} \left(1 - \frac{x}{2\pi}\right)^{n-1} dz = \frac{2\pi}{n}.$$

   Let us define the indicator random variable $X_{u,s}$ that takes on the value 1 iff $h(u) = s$. Crucially, we observe that a user $u \in \mathcal{U}$ maps to the server $h(u) = s \in \mathcal{S}$ iff $\theta_u \in Z_s$. Observe that since θ_u is chosen uniformly at random and independently of $\{\theta_s\}_{s\in\mathcal{S}}$, we have

$$\mathbb{E}[X_{u,s}] = \mathbb{E}_s[\mathbb{E}_u[X_{u,s}]] = \mathbb{E}_s[Z_s/(2\pi)] = 1/n.$$

   Finally, let $X_s = \sum_{u\in\mathcal{U}} X_{u,s}$ be the number of users assigned to the server s. By linearity of expectation, we conclude that $\mathbb{E}[X_s] = m/n$.

2. Now, suppose that we introduce a new server s'. We select a new uniformly random point θ' for s'. Without loss of generality, suppose that θ' lies in Z_s for some $s \in \mathcal{S}$. Notice that θ' partitions Z_s into two arcs and the only users that we have to reassign to s' are the ones that lie precisely in the arc from this partition that lies clockwise starting from θ'. Thus, the number of such users is clearly upper bounded by the number of users previously assigned to s, which is m/n in expectation.

3. We now have $Z_s = \frac{2\pi}{n}$ deterministically for every $s \in \mathcal{S}$. Thus, for every u, s, $X_{u,s}$ is a Bernoulli random variable with a success probability of $1/n$. Hence, when $\{\theta_u\}_{u\in\mathcal{U}}$ are fully independent, X_s is a sum of m independent and identically distributed Bernoulli random variables. Consequently, by Chernoff's Bounds, we have

$$\mathbb{P}[X_s \geq (1+\varepsilon)\mathbb{E}[X_s]] \leq e^{-\frac{\varepsilon^2\mathbb{E}[X_s]}{2+\varepsilon}} = e^{-\frac{\varepsilon^2(m/n)}{2+\varepsilon}}.$$

   Now, we use the fact that $m = \omega(n \log^2 n)$ to conclude that this probability is $n^{-\omega(1)}$. Finally, we can union bound over all servers to prove the required claim.

4. Notice that X_s is now a sum of m Bernoulli random variables that are k-wise independent. Since the random variables are k-wise independent, they are also $\lfloor\varepsilon k\rfloor$-wise independent. Additionally, for a sufficiently large n, we have $\lfloor\varepsilon k\rfloor \geq 2$. Thus, we can now apply Theorem 1.10 from the script to obtain

$$\mathbb{P}[|X_s - \mathbb{E}[X_s]| \geq \varepsilon\mathbb{E}[X_s]] = \mathcal{O}\left(\left(\frac{\lfloor\varepsilon k\rfloor}{2\varepsilon\sqrt{\mathbb{E}[X_s]}}\right)^{\lfloor\varepsilon k\rfloor}\right) = \mathcal{O}\left(\left(\frac{k}{2\sqrt{m/n}}\right)^{\lfloor\varepsilon k\rfloor}\right).$$

Now, we use the fact that $k = \lfloor \log n \rfloor$ and $m = \omega(n \log^2 n)$ to conclude that this probability is $n^{-\omega(1)}$. Finally, we can union bound over all servers to prove the required claim.

2 K-means+++

Raghu Raman Ravi

Let the optimal clustering of the set of given points X be $\{A_i^*\}_{i\in[k]}$. Additionally, let $\phi_i^* = \phi(A_i^*, \mu_{A_i^*})$ for all $i \in [k]$, and let $\phi^* = \sum_{i\in[k]} \phi_i^*$ be the optimal cost.

Let the cluster center sampled at time step t of the K-means+++ algorithm be c_t and let $C_t = \{c_t \mid 1 \le i \le t\}$. As in the lecture, at the end of the time step t we define an optimal cluster A_i^* to be *settled* iff we have $\phi(A_i^*, C_t) \le 10\phi_i^*$. Additionally, we define $S_t = \{i \mid \phi(A_i^*, C_t) \le 10\phi_i^*\}$ as the indices of the settled clusters at the end of the time step t.

First, we will show that every step, if we do not have a good approximation yet, we sample from an unsettled cluster with constant probability. Assume that at the end of t time steps, we have $\phi(X, C_t) > 20\phi^*$. Now, notice that

$$\sum_{i\in S_t} \phi(A_i^*, C_t) \le \sum_{i\in S_t} 10\phi_i^* \le \sum_{i\in[k]} 10\phi_i^* \le 10\phi^*.$$

Let $E_{t,i}$ be the event that we sample from the cluster A_i^* at time step t and let $E_t := \cup_{i\in S_t} E_{t,i}$ be the event that we sample from an unsettled cluster in time step t. Then, we have

$$\mathbb{P}[E_{t+1}|C_t] = \sum_{i\notin S_t} \mathbb{P}[E_{t+1,i}|C_t] = \frac{\sum_{i\notin S_t} \phi(A_i^*, C_t)}{\phi(X, C_t)} = 1 - \frac{\sum_{i\in S_t} \phi(A_i^*, C_t)}{\phi(X, C_t)} \ge 1 - \frac{10\phi^*}{20\phi^*} = \frac{1}{2}.$$

Now, suppose that at a time step $t+1$ we sample the center c_{t+1} from an unsettled cluster A_i^*. If $t \ge 2$, by applying Lemma 9.3 from the lecture notes, we conclude that $\mathbb{E}[\phi(A_i^*, C_{t+1}\}) \mid C_t, E_{t+1,i}] \le 8\phi_i^*$. If $t = 1$, the same claim still holds by applying Lemma 9.2 from the lecture notes. Now, let $S_{t+1,i}$ be the event that we settle the previously unsettled cluster i at the end of the time step $t+1$ and also sample c_{t+1} from this cluster. By Markov's inequality we conclude that $\mathbb{P}[S_{t+1,i} \mid C_t, E_{t+1,i}] \ge 1 - \frac{8\phi_i^*}{10\phi_i^*} = \frac{1}{5}$. Thus, for an unsettled cluster i, we have $\mathbb{P}[S_{t+1,i} \mid C_t] \ge \mathbb{P}[E_{t+1,i} \mid C_t]/5$.

Now, let F_t be the event that we settle some unsettled cluster in the time step t. From the previous claims, we conclude that

$$\mathbb{P}[F_t \mid C_t] \ge \sum_{i\notin S} \mathbb{P}[S_{t,i} \mid C_t] \ge \sum_{i\notin S} \frac{\mathbb{P}[E_{t,i} \mid C_t]}{5} \ge \frac{\mathbb{P}[E_t \mid C_t]}{5} \ge \frac{1}{10}.$$

Notice that this is an uniform bound for any instantiation of C_t. Taking an expectation over C_t, we now obtain $\mathbb{P}[F_t] \ge 1/10$.

Finally, we define for every time step t, the random variable $X_t = \mathbf{1}\{F_t\}$ to be the indicator random variable for F_t. Using the Azuma-Hoeffding bound with $\delta = 1/2$, we conclude that

$$\mathbb{P}\left[\sum_{1\le i\le 100k} X_t \le 5k\right] \le \exp\left(-\frac{\left(\frac{1}{2}\right)^2 \left(\frac{1}{10}\right)^2 (100k)}{2}\right) \le e^{-k/8},$$

which immediately implies out required claim.

3 High-Coverage Forest

Antti Roeyskoe

1. Fix some optimum solution T^*, which is a forest consisting of up to k vertex-disjoint subtrees. Denote by V^* the set of vertices that can reach a center in T^*.

For a subtree T_j^* of T^* with a center, let P_j^* be its Euler tour, i.e. a non-simple path that traverses every edge of T^* exactly once in each direction. Then, $w(P_{T_j^*}) = 2w(T_j^*)$.

Partition these paths $P_1^*, \ldots, P_k^*$ into maximal segments $P_1', \ldots, P_{k'}'$ each with weight at most B/t. Note that some of P_j' might consist of a single vertex and no edges, if the following edge on the corresponding Euler tour had weight greater than B/t. Since the total weight of the paths $P_1^*, \ldots, P_k^*$ is at most $2B$, the number of segments k' satisfies $k' \leq k + \lfloor 2B/(B/t) \rfloor = k + 2t$.

Now, greedily group these segments into groups, each containing at most k segments and having weight at most B/t. A new group has to be started because of the first condition at most $\lfloor (k+2t)/k \rfloor \leq t$ times, and a new group has to be started because of the second condition at most $4t$ times. Thus, the number of groups is $O(t)$. Since each vertex in V^* appears in one of the groups, the best group is a B/t-cost solution with value at least $v(V^*)/O(t) = \Omega(\mathrm{OPT}(G, B, k)/t)$.

2. We first round the values, which is the only source of approximation. Since it is always possible to take the highest-value vertex as a center, we have $\mathrm{OPT}(G, B/t, k) \geq \mathrm{OPT}(G, 0, 1) \geq \max_v r(v)$. Let $r'(v) := \lfloor nr(v)/\max_v r(v) \rfloor$. The optimal solution with these values is constant-approximate to the original optimum.

   Next, root the tree at an arbitrary vertex, and note that we may assume that each vertex in the tree is either a leaf or has exactly two children, by adding edges of weight 0 and vertices of value 0 to split high-degree vertices.

   We will now perform tree DP, computing for each subtree (rooted at vertex v with children $c_1, \ldots, c_l$) a dp table $\mathrm{DP}[v][b][x][y]$, giving the minimum total weight of edges that need to be included into T in the subtree, given that we place x centers into the subtree, vertex v can reach one of them if $b = 1$ or otherwise cannot if $b = 0$, and the total number of vertices in the subtree that can reach a center is y. The DP table will additionally store the edges the corresponding solution uses.

   For a leaf vertex v, we have $\mathrm{DP}[v][0][0][0] = 0$, $\mathrm{DP}[v][1][1][1] = 0$, and $\mathrm{DP}[v][b][x][y] = \infty$ for all other b, x, y. For a non-leaf vertex v with two children v_1 and v_2 to which the edges have weights w_1 and w_2, we iterate over all combinations of $b_1, b_2, x_1, x_2, y_1, y_2$, of which there are $O(k^2 n^4)$ as the sum of values of each vertex is at most $O(n^2)$. Let $w_{\mathrm{sub}} := \mathrm{DP}[v_1][b_1][x_1][y_1] + \mathrm{DP}[v_2][b_2][x_2][y_2]$. Then,

   - If vertex v doesn't need to reach a center, i.e. $b = 0$, we get the offer $\mathrm{DP}[v][0][x_1 + x_2][y_1 + y_2] \leq w_{\mathrm{sub}}$.
   - Otherwise, if $b = 1$ and neither of the children needs to reach a center, i.e. $b_1 = b_2 = 0$, we get the offer $\mathrm{DP}[v][1][x_1 + x_2 + 1][y_1 + y_2 + r'(v)] \leq w_{\mathrm{sub}}$.
   - Otherwise, if $b = 1$ and WLOG $b_1 = 1$, $b_2 = 0$, then note that we do not need to add a new center, and we get the offer $\mathrm{DP}[v][1][x_1 + x_2][y_1 + y_2 + r'(v)] \leq w_{\mathrm{sub}} + w_1$.
   - Otherwise, if $b = b_1 = b_2 = 1$, then note that we can in fact remove one of the centers, and we get the offer $\mathrm{DP}[v][1][x_1 + x_2 - 1][y_1 + y_2 + r'(v)] \leq w_{\mathrm{sub}} + w_1 + w_2$.

   We then simply set the DP value to the minimum upper bound obtained, or ∞ if there was none or the resulting index is out of bounds. The corresponding solution uses the union of the two subtrees' corresponding solutions, plus the edge to child v_1 if we pay w_1 and the edge to child v_2 if we pay w_2.

   Finally, we can return the solution corresponding to $\mathrm{DP}[\mathrm{root}][b][k'][a]$ for the maximum a for which there are some b and $k' \leq k$ for which the DP value is at most the budget.

3. Recall the low-stretch tree embedding result from the lectures:

   Let $\mathcal{Q}$ be a tree embedding distribution for the graph G with a stretch at most $\alpha = \mathcal{O}(\log n)$. Suppose that Q is sampled from $\mathcal{Q}$. We assign a value of zero to all auxiliary vertices that we create in Q. Let T' be the optimal solution to the problem on Q with a budget of B. We obtain a new solution T in G by mapping all the edges in T' to their corresponding paths in G. Additionally, let T^* be the optimal solution to the original problem in G with a budget of $B/(2\alpha)$. Finally, let T'' be the solution in Q obtained by mapping the edges in T^* to Q.

   Notice that $\mathbb{E}[w(T'')] \leq \alpha w(T^*) \leq B/2$. By Markov's Inequality, with probability at least $1/2$, we have $w(T'') \leq B$. Thus, we have $OPT(Q, B, k) \geq r(T'') = r(T^*)$. Now, by the optimality of T^*, we have $r(T^*) = OPT(G, B/2\alpha, k) = \Omega(OPT(G, B, k)/\alpha)$ (By using part (1)).

   Now, notice that $w(T) \leq w(T') \leq B$. Additionally, we have $r(T) = r(T') \geq OPT(Q, B, k)$ (By using part (2)). But, we have $OPT(Q, B, k) \geq r(T^*) = \Omega(OPT(G, B, k)/\alpha)$. Hence, we must have $r(T) = \Omega(OPT(G, B, k)/\alpha)$. Hence, T must be an $\mathcal{O}(\alpha) = \mathcal{O}(\log n)$ approximation with a probability of at least $1/2$. We can now just repeat the process $\mathcal{O}(\log n)$ times to boost the success probability to $1 - n^{-\Omega(1)}$.

4 Flow from Short Paths

Antti Roeyskoe

1. In each iteration, there is some edge e_i for which $u_i^{\min} = u(e_i)$. The length of that edge increases by a multiplicative $1+\epsilon$. Since each edge has initial length δ and edges of length 1 or greater do not appear on any path P_i, the number of iterations is at most

$$m \cdot \lceil \log_{1+\epsilon}(1/\delta) \rceil = O\left(m \cdot \frac{\ln(1/\delta)}{\ln(1+\epsilon)}\right) = O\left(m \cdot \frac{\ln(1/\delta)}{\epsilon}\right)$$

   where we used $\ln(1+x) \geq \frac{x}{1+x}$ for $x > -1$.

2. As argued in subtask 1, we have $\ell(e) \leq (1+\epsilon)$ for each edge e. On the other hand, using the inequality $1 + x \geq e^{x-x^2} \geq e^{x-\epsilon x} = e^{(1-\epsilon)x}$ for $0 \leq x \leq \epsilon$, we get

$$\begin{aligned} \ell_i(e) &= \ell_{i-1}(e) \cdot (1 + (f_i(e) - f_{i-1}(e))/u(e)) \\ &\geq \ell_{i-1}(e) \cdot e^{(1-\epsilon)(f_i(e)-f_{i-1}(e))/u(e)}, \end{aligned}$$

   thus in particular, $\ell(e) \geq \delta \cdot \exp((1-\epsilon)f(e)/u(e))$. Now, applying the upper bound $1+\epsilon$ to the left side and taking logarithms of both sides, we get that

$$\ln(1+\epsilon) \geq \ln(\delta) + (1-\epsilon)\frac{f(e)}{u(e)}$$

   and by moving terms,

$$\frac{f(e)}{u(e)} \leq \frac{\ln(1+\epsilon) - \ln(\delta)}{1-\epsilon} = \frac{\ln \frac{1}{\delta}}{1-\epsilon} + O(1).$$

3. Since the lengths $\ell_i - \ell_0$ are nonnegative, any shortest s-t -path P in G with edge lengths $\ell_i - \ell_0$ has at most $n-1 \leq n$ edges. Thus, it has length at least $\ell_i(P) - \ell_0(P) \geq \ell_i(P) - \delta n \geq d_i - \delta n$. Thus, scaling the lengths $\ell_i - \ell_0$ by $\frac{1}{d_i - \delta n}$, any s-t -path will have length at least 1. Then, since the scaled lengths are a valid solution to the dual, $\langle \frac{1}{d_i - \delta n}(\ell_i - \ell_0), u \rangle \geq \beta$ must hold. This indeed is equivalent to

$$\langle \ell_i, u \rangle - \langle \ell_0, u \rangle \geq (d_i - \delta n)\beta. \tag{1}$$

4. Since the flow only changes on P_i, and on those edges increases by $\text{val}(f_i) - \text{val}(f_{i-1})$,

$$\begin{aligned}\langle \ell_i, u\rangle &= \langle \ell_{i-1}, u\rangle + \sum_{e\in P_i} \ell_{i-1}(e)\cdot(\text{val}(f_i) - \text{val}(f_{i-1}))\\ &= \langle \ell_{i-1}, u\rangle + d_{i-1}\cdot(\text{val}(f_i) - \text{val}(f_{i-1}).\end{aligned}$$

Applying this repeatedly, we get

$$\langle \ell_i, u\rangle - \langle \ell_0, u\rangle = \sum_{j=1}^{i} d_{j-1}\cdot(\text{val}(f_j) - \text{val}(f_{j-1})). \tag{2}$$

Combining Equation (1) and Equation (2), we get

$$(d_i - \delta n)\beta \le \sum_{j=1}^{i} d_{j-1}\cdot(\text{val}(f_j) - \text{val}(f_{j-1}))$$

and by moving terms,

$$d_i \le \delta n + \frac{1}{\beta}\sum_{j=1}^{i} d_{j-1}\cdot(\text{val}(f_j) - \text{val}(f_{j-1})).$$

This upper bound on d_i is maximized when each of the values $d_1, \ldots, d_{i-1}$ are also maximized, i.e. the above equality holds for them. In particular, we can define a sequence $h_0, h_1, \ldots, h_t$ satisfying $d_i \le h_i$ by letting $h_0 := \delta n$ and

$$h_i = \delta n + \frac{1}{\beta}\sum_{j=1}^{i} h_{j-1}\cdot(\text{val}(f_j) - \text{val}(f_{j-1})).$$

Then, we have

$$\begin{aligned}h_i &= h_{i-1} + h_{i-1}\frac{\text{val}(f_i) - \text{val}(f_{i-1})}{\beta}\\ &= h_{i-1}\left(1 + \frac{\text{val}(f_i) - \text{val}(f_{i-1})}{\beta}\right)\\ &\le h_{i-1}\cdot\exp\left(\frac{\text{val}(f_i) - \text{val}(f_{i-1})}{\beta}\right),\end{aligned}$$

thus, as desired, we have

$$d_i \le h_i \le \delta n\cdot\exp\left(\frac{\text{val}(f_i)}{\beta}\right).$$

Setting $i = k$ and noting that $d_t \ge 1/(1+\epsilon)$ as $d_{t+1} \ge 1$ and the length of the path P_k increases by at most a multiplicative $(1+\epsilon)$ in iteration k, and then taking logarithms of both sides, we obtain

$$\ln(1/(1+\epsilon)) \le \ln(d_k) \le \ln(\delta n) + \frac{\text{val}(f)}{\beta},$$

and by moving terms that

$$\text{val}(f) \ge \beta\cdot\left(\ln(\frac{1}{\delta n}) - O(1)\right). \tag{3}$$

5. Let f' be f scaled down by the appropriate term $\frac{\ln \frac{1}{\delta}}{1-\epsilon} + O(1)$ to be capacity-respecting, as we obtained in part 2. Then, using part 4 to lower bound $\text{val}(f)$, we have

$$\text{val}(f') := \text{val}(f) \cdot (1-\epsilon) \left(\frac{1}{\ln \frac{1}{\delta} + O(1)} \right) \geq \beta \cdot (1-\epsilon) \left(\frac{\ln \frac{1}{\delta n} - O(1)}{\ln \frac{1}{\delta} + O(1)} \right).$$

Now, pick $\delta = n^{-1/\epsilon}$, making the right hand side equal to

$$\beta(1-\epsilon) \left(\frac{\ln \frac{1}{\delta n} - O(1)}{\ln \frac{1}{\delta} + O(1)} \right) \geq \beta(1-\epsilon) \left(\frac{(\frac{1}{\epsilon} - O(1)) \ln(n)}{(\frac{1}{\epsilon} + O(1)) \ln n} \right) \geq \beta(1-\epsilon)(1-O(\epsilon)) = \beta(1-O(\epsilon))$$

for ϵ at most a sufficiently small constant. Since β is the value of the optimum flow, the flow f' is thus $(1 - O(\epsilon))$-approximate.

Plugging this δ into the iteration bound of part 1, we get that the number of iterations is at most $O(\frac{m \ln n}{\epsilon^2})$ as desired.